Solutions to Section 7.4

Trigonometric Substitutions

MAT120 Course Materials

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Summary of Key Trigonometric Substitutions

Radical / Expression	Substitution	Differential	Identity Used
$\sqrt{a^2 - x^2}$	$x = a\sin\theta$	$dx = a\cos\theta\, d\theta$	$1 - \sin^2\theta = \cos^2\theta$
$\sqrt{a^2 + x^2}$	$x = a\tan\theta$	$dx = a\sec^2\theta\, d\theta$	$1 + \tan^2\theta = \sec^2\theta$
$\sqrt{x^2 - a^2}$	$x = a\sec\theta$	$dx = a\sec\theta\tan\theta\, d\theta$	$\sec^2\theta - 1 = \tan^2\theta$

Part I: Quick Check Exercises 7.4

1. Give a trigonometric substitution that will eliminate the radical:

(a) $\sqrt{a^2 - x^2} \implies \mathbf{x = a\sin\theta}$ (or $x = a\cos\theta$)

(b) $\sqrt{a^2 + x^2} \implies \mathbf{x = a\tan\theta}$

(c) $\sqrt{x^2 - a^2} \implies \mathbf{x = a\sec\theta}$

2. If $x = 2\sec\theta$ and $0 < \theta < \pi/2$, then:

(a) $\sin\theta = \frac{\sqrt{\mathbf{x^2-4}}}{\mathbf{x}}$

(b) $\cos\theta = \frac{\mathbf{2}}{\mathbf{x}}$

(c) $\tan\theta = \frac{\sqrt{\mathbf{x^2-4}}}{\mathbf{2}}$

3. State the trigonometric substitution to eliminate the radical (do not evaluate):

(a) $\int \sqrt{9 + x^2}\, dx \implies \mathbf{x = 3\tan\theta}$

(b) $\int \sqrt{9 - x^2}\, dx \implies \mathbf{x = 3\sin\theta}$

(c) $\int \sqrt{1 - 9x^2}\, dx \implies \mathbf{x = \frac{1}{3}\sin\theta}$

4. Determine the substitution u for completing the square:

(a) $\int \frac{1}{x^2-2x+10}\, dx = \int \frac{1}{(x-1)^2+3^2}\, dx \implies \mathbf{u = x - 1}$

(b) $\int \sqrt{x^2 - 6x + 8}\, dx = \int \sqrt{(x-3)^2 - 1}\, dx \implies \mathbf{u = x - 3}$

(c) $\int \sqrt{12 - 4x - x^2}\, dx = \int \sqrt{16 - (x+2)^2}\, dx \implies \mathbf{u = x + 2}$

Part II: Exercise Set 7.4 (Problems 1–26)

Indefinite Integrals

1. Evaluate $\int \sqrt{9-x^2}\,dx$:

Let $x = 3\sin\theta$, so $dx = 3\cos\theta\,d\theta$ and $\sqrt{9-x^2} = 3\cos\theta$.

$$\int 3\cos\theta \cdot 3\cos\theta\,d\theta = 9\int \cos^2\theta\,d\theta = \frac{9}{2}\int (1+\cos 2\theta)\,d\theta = \frac{9}{2}\theta + \frac{9}{4}\sin 2\theta + C$$

Using $\sin 2\theta = 2\sin\theta\cos\theta = 2\left(\frac{x}{3}\right)\left(\frac{\sqrt{9-x^2}}{3}\right) = \frac{2x\sqrt{9-x^2}}{9}$ and $\theta = \arcsin\left(\frac{x}{3}\right)$:

$$= \frac{\mathbf{9}}{\mathbf{2}}\arcsin\left(\frac{\mathbf{x}}{\mathbf{3}}\right) + \frac{\mathbf{x}\sqrt{\mathbf{9-x^2}}}{\mathbf{2}} + \mathbf{C}$$

2. Evaluate $\int \sqrt{1-4x^2}\,dx$:

Let $2x = \sin\theta \implies x = \frac{1}{2}\sin\theta$, $dx = \frac{1}{2}\cos\theta\,d\theta$, and $\sqrt{1-4x^2} = \cos\theta$.

$$\int \cos\theta\left(\frac{1}{2}\cos\theta\right)d\theta = \frac{1}{2}\int \cos^2\theta\,d\theta = \frac{1}{4}\theta + \frac{1}{8}\sin 2\theta + C$$

$$= \frac{\mathbf{1}}{\mathbf{4}}\arcsin(\mathbf{2x}) + \frac{\mathbf{x}\sqrt{\mathbf{1-4x^2}}}{\mathbf{2}} + \mathbf{C}$$

3. Evaluate $\int \frac{x^2}{\sqrt{4-x^2}}\,dx$:

Let $x = 2\sin\theta$, $dx = 2\cos\theta\,d\theta$, $\sqrt{4-x^2} = 2\cos\theta$.

$$\int \frac{4\sin^2\theta}{2\cos\theta}(2\cos\theta\,d\theta) = 4\int \sin^2\theta\,d\theta = 2\theta - \sin 2\theta + C$$

$$= \mathbf{2}\arcsin\left(\frac{\mathbf{x}}{\mathbf{2}}\right) - \frac{\mathbf{x}\sqrt{\mathbf{4-x^2}}}{\mathbf{2}} + \mathbf{C}$$

4. Evaluate $\int \frac{dx}{x^2\sqrt{9-x^2}}$:

Let $x = 3\sin\theta$, $dx = 3\cos\theta\,d\theta$, $\sqrt{9-x^2} = 3\cos\theta$.

$$\int \frac{3\cos\theta\,d\theta}{9\sin^2\theta\cdot 3\cos\theta} = \frac{1}{9}\int \csc^2\theta\,d\theta = -\frac{1}{9}\cot\theta + C$$

Since $\cot\theta = \frac{\sqrt{9-x^2}}{x}$:

$$= -\frac{\sqrt{\mathbf{9-x^2}}}{\mathbf{9x}} + \mathbf{C}$$

5. Evaluate $\int \frac{dx}{(4+x^2)^2}$:

Let $x = 2\tan\theta$, $dx = 2\sec^2\theta\,d\theta$, $4+x^2 = 4\sec^2\theta$.

$$\int \frac{2\sec^2\theta\,d\theta}{16\sec^4\theta} = \frac{1}{8}\int \cos^2\theta\,d\theta = \frac{1}{16}\theta + \frac{1}{32}\sin 2\theta + C$$

Since $\sin 2\theta = \frac{2x}{x^2+4}$:

$$= \frac{\mathbf{1}}{\mathbf{16}}\arctan\left(\frac{\mathbf{x}}{\mathbf{2}}\right) + \frac{\mathbf{x}}{\mathbf{8(x^2+4)}} + \mathbf{C}$$

6. Evaluate $\displaystyle\int \frac{x^2}{\sqrt{7+x^2}}\,dx$**:**

Let $x = \sqrt{7}\tan\theta$, $dx = \sqrt{7}\sec^2\theta\,d\theta$, $\sqrt{7+x^2} = \sqrt{7}\sec\theta$.

$$\int \frac{7\tan^2\theta}{\sqrt{7}\sec\theta}(\sqrt{7}\sec^2\theta\,d\theta) = 7\int \tan^2\theta\sec\theta\,d\theta = 7\int(\sec^3\theta - \sec\theta)\,d\theta$$

Using $\int \sec^3\theta\,d\theta = \frac{1}{2}\sec\theta\tan\theta + \frac{1}{2}\ln|\sec\theta+\tan\theta|$:

$$= \frac{7}{2}\sec\theta\tan\theta - \frac{7}{2}\ln|\sec\theta+\tan\theta| + C$$

Substituting back $\tan\theta = \frac{x}{\sqrt{7}}$ and $\sec\theta = \frac{\sqrt{x^2+7}}{\sqrt{7}}$:

$$= \frac{\mathbf{x\sqrt{x^2+7}}}{\mathbf{2}} - \frac{\mathbf{7}}{\mathbf{2}}\ln\left|\mathbf{x} + \sqrt{\mathbf{x^2+7}}\right| + \mathbf{C'}$$

7. Evaluate $\displaystyle\int \frac{\sqrt{x^2-9}}{x}\,dx$**:**

Let $x = 3\sec\theta$, $dx = 3\sec\theta\tan\theta\,d\theta$, $\sqrt{x^2-9} = 3\tan\theta$.

$$\int \frac{3\tan\theta}{3\sec\theta}(3\sec\theta\tan\theta\,d\theta) = 3\int\tan^2\theta\,d\theta = 3\int(\sec^2\theta - 1)\,d\theta = 3\tan\theta - 3\theta + C$$

$$= \sqrt{\mathbf{x^2-9}} - \mathbf{3}\text{arcsec}\left(\frac{\mathbf{x}}{\mathbf{3}}\right) + \mathbf{C} \quad \left[\text{or } \sqrt{\mathbf{x^2-9}} - \mathbf{3}\arctan\left(\frac{\sqrt{\mathbf{x^2-9}}}{\mathbf{3}}\right) + \mathbf{C}\right]$$

8. Evaluate $\displaystyle\int \frac{dx}{x^2\sqrt{x^2-4}}$**:**

Let $x = 2\sec\theta$, $dx = 2\sec\theta\tan\theta\,d\theta$, $\sqrt{x^2-4} = 2\tan\theta$.

$$\int \frac{2\sec\theta\tan\theta\,d\theta}{4\sec^2\theta\cdot 2\tan\theta} = \frac{1}{4}\int\cos\theta\,d\theta = \frac{1}{4}\sin\theta + C$$

$$= \frac{\sqrt{\mathbf{x^2-4}}}{\mathbf{4x}} + \mathbf{C}$$

9. Evaluate $\displaystyle\int \frac{3x^3}{\sqrt{1-x^2}}\,dx$**:**

Let $u = 1-x^2 \implies x^2 = 1-u$, $du = -2x\,dx \implies x\,dx = -\frac{du}{2}$.

$$\int \frac{3(1-u)}{\sqrt{u}}\left(-\frac{du}{2}\right) = -\frac{3}{2}\int(u^{-1/2} - u^{1/2})\,du = -\frac{3}{2}\left(2u^{1/2} - \frac{2}{3}u^{3/2}\right) + C$$

$$= u^{3/2} - 3u^{1/2} + C = \sqrt{1-x^2}(1-x^2-3) + C = -(\mathbf{x^2+2})\sqrt{\mathbf{1-x^2}} + \mathbf{C}$$

10. Evaluate $\displaystyle\int x^3\sqrt{5-x^2}\,dx$**:**

Let $u = 5-x^2 \implies x^2 = 5-u$, $du = -2x\,dx$.

$$\int(5-u)\sqrt{u}\left(-\frac{du}{2}\right) = \frac{1}{2}\int(u^{3/2} - 5u^{1/2})\,du = \frac{1}{2}\left(\frac{2}{5}u^{5/2} - 5\cdot\frac{2}{3}u^{3/2}\right) + C$$

$$= \frac{1}{5}(5-x^2)^{5/2} - \frac{5}{3}(5-x^2)^{3/2} + C = \frac{\sqrt{\mathbf{5-x^2}}}{\mathbf{15}}(\mathbf{3x^4 - 5x^2 - 50}) + \mathbf{C}$$

11. Evaluate $\displaystyle\int \frac{dx}{x^2\sqrt{9x^2-4}}$**:**

Let $3x = 2\sec\theta \implies x = \frac{2}{3}\sec\theta$, $dx = \frac{2}{3}\sec\theta\tan\theta\,d\theta$, $\sqrt{9x^2-4} = 2\tan\theta$.

$$\int \frac{\frac{2}{3}\sec\theta\tan\theta\,d\theta}{\frac{4}{9}\sec^2\theta\cdot 2\tan\theta} = \frac{3}{4}\int \cos\theta\,d\theta = \frac{3}{4}\sin\theta + C$$

Since $\sin\theta = \frac{\sqrt{9x^2-4}}{3x}$:

$$= \frac{\sqrt{\mathbf{9x^2-4}}}{\mathbf{4x}} + \mathbf{C}$$

12. Evaluate $\displaystyle\int \frac{\sqrt{1+t^2}}{t}\,dt$**:**

Let $t = \tan\theta$, $dt = \sec^2\theta\,d\theta$, $\sqrt{1+t^2} = \sec\theta$.

$$\int \frac{\sec\theta}{\tan\theta}\sec^2\theta\,d\theta = \int \frac{\sec\theta(1+\tan^2\theta)}{\tan\theta}\,d\theta = \int (\csc\theta + \sec\theta\tan\theta)\,d\theta$$

$$= \sec\theta - \ln|\csc\theta+\cot\theta| + C = \sqrt{\mathbf{1+t^2}} - \ln\left|\frac{\mathbf{1+\sqrt{1+t^2}}}{\mathbf{t}}\right| + \mathbf{C}$$

13. Evaluate $\displaystyle\int \frac{dx}{(1-x^2)^{3/2}}$**:**

Let $x = \sin\theta$, $dx = \cos\theta\,d\theta$, $(1-x^2)^{3/2} = \cos^3\theta$.

$$\int \frac{\cos\theta\,d\theta}{\cos^3\theta} = \int \sec^2\theta\,d\theta = \tan\theta + C = \frac{\mathbf{x}}{\sqrt{\mathbf{1-x^2}}} + \mathbf{C}$$

14. Evaluate $\displaystyle\int \frac{dx}{x^2\sqrt{x^2+25}}$**:**

Let $x = 5\tan\theta$, $dx = 5\sec^2\theta\,d\theta$, $\sqrt{x^2+25} = 5\sec\theta$.

$$\int \frac{5\sec^2\theta\,d\theta}{25\tan^2\theta\cdot 5\sec\theta} = \frac{1}{25}\int \csc^2\theta\,d\theta = -\frac{1}{25}\cot\theta + C = -\frac{\sqrt{\mathbf{x^2+25}}}{\mathbf{25x}} + \mathbf{C}$$

15. Evaluate $\displaystyle\int \frac{dx}{\sqrt{x^2-16}}$**:**

Let $x = 4\sec\theta$, $dx = 4\sec\theta\tan\theta\,d\theta$, $\sqrt{x^2-16} = 4\tan\theta$.

$$\int \frac{4\sec\theta\tan\theta\,d\theta}{4\tan\theta} = \int \sec\theta\,d\theta = \ln|\sec\theta+\tan\theta| + C = \ln\left|\mathbf{x}+\sqrt{\mathbf{x^2-16}}\right| + \mathbf{C}'$$

16. Evaluate $\displaystyle\int \frac{dx}{1+2x^2+x^4} = \int \frac{dx}{(1+x^2)^2}$**:**

Let $x = \tan\theta$, $dx = \sec^2\theta\,d\theta$, $1+x^2 = \sec^2\theta$.

$$\int \frac{\sec^2\theta\,d\theta}{\sec^4\theta} = \int \cos^2\theta\,d\theta = \frac{1}{2}\theta + \frac{1}{4}\sin 2\theta + C = \frac{\mathbf{1}}{\mathbf{2}}\arctan(\mathbf{x}) + \frac{\mathbf{x}}{\mathbf{2(1+x^2)}} + \mathbf{C}$$

17. Evaluate $\displaystyle\int \frac{dx}{(4x^2-9)^{3/2}}$**:**

Let $2x = 3\sec\theta \implies x = \frac{3}{2}\sec\theta$, $dx = \frac{3}{2}\sec\theta\tan\theta\,d\theta$, $(4x^2-9)^{3/2} = 27\tan^3\theta$.

$$\int \frac{\frac{3}{2}\sec\theta\tan\theta\,d\theta}{27\tan^3\theta} = \frac{1}{18}\int \frac{\sec\theta}{\tan^2\theta}\,d\theta = \frac{1}{18}\int \frac{\cos\theta}{\sin^2\theta}\,d\theta = -\frac{1}{18\sin\theta} + C$$

$$= -\frac{\mathbf{x}}{\mathbf{9\sqrt{4x^2-9}}} + \mathbf{C}$$

18. Evaluate $\displaystyle\int \frac{3x^3}{\sqrt{x^2-25}}\,dx$**:**

Let $u = x^2 - 25 \implies x^2 = u + 25$, $du = 2x\,dx$.

$$\int \frac{3(u+25)}{2\sqrt{u}}\,du = \frac{3}{2}\int (u^{1/2} + 25u^{-1/2})\,du = \frac{3}{2}\left(\frac{2}{3}u^{3/2} + 50u^{1/2}\right) + C$$

$$= u^{3/2} + 75u^{1/2} + C = \sqrt{x^2-25}(x^2 - 25 + 75) + C = \mathbf{(x^2 + 50)\sqrt{x^2 - 25} + C}$$

19. Evaluate $\displaystyle\int e^x\sqrt{1-e^{2x}}\,dx$**:**

Let $u = e^x \implies du = e^x\,dx$. The integral becomes $\int \sqrt{1-u^2}\,du$.

$$= \frac{u\sqrt{1-u^2}}{2} + \frac{1}{2}\arcsin(u) + C = \mathbf{\frac{e^x\sqrt{1-e^{2x}}}{2} + \frac{1}{2}\arcsin(e^x) + C}$$

20. Evaluate $\displaystyle\int \frac{\cos\theta}{\sqrt{2-\sin^2\theta}}\,d\theta$**:**

Let $u = \sin\theta \implies du = \cos\theta\,d\theta$. The integral becomes $\int \frac{du}{\sqrt{2-u^2}}$.

$$= \arcsin\left(\frac{u}{\sqrt{2}}\right) + C = \arcsin\left(\frac{\sin\theta}{\sqrt{2}}\right) + \mathbf{C}$$

Definite Integrals

21. Evaluate $\displaystyle\int_0^1 5x^3\sqrt{1-x^2}\,dx$**:**

Let $u = 1-x^2 \implies du = -2x\,dx$. Limits: $x=0 \implies u=1$; $x=1 \implies u=0$.

$$\int_1^0 5(1-u)\sqrt{u}\left(-\frac{du}{2}\right) = \frac{5}{2}\int_0^1 (u^{1/2}-u^{3/2})\,du = \frac{5}{2}\left[\frac{2}{3}u^{3/2}-\frac{2}{5}u^{5/2}\right]_0^1 = \frac{5}{2}\left(\frac{2}{3}-\frac{2}{5}\right) = \mathbf{\frac{2}{3}}$$

22. Evaluate $\displaystyle\int_0^{1/2}\frac{dx}{(1-x^2)^2}$**:**

Let $x = \sin\theta \implies dx = \cos\theta\,d\theta$. Limits: $x=0 \implies \theta=0$; $x=1/2 \implies \theta=\pi/6$.

$$\int_0^{\pi/6}\frac{\cos\theta\,d\theta}{\cos^4\theta} = \int_0^{\pi/6}\sec^3\theta\,d\theta = \frac{1}{2}\left[\sec\theta\tan\theta + \ln|\sec\theta+\tan\theta|\right]_0^{\pi/6}$$

At $\theta=\pi/6$: $\sec(\pi/6)=\frac{2}{\sqrt{3}}$, $\tan(\pi/6)=\frac{1}{\sqrt{3}}$.

$$= \frac{1}{2}\left(\frac{2}{3}+\ln\left(\frac{3}{\sqrt{3}}\right)\right) = \mathbf{\frac{1}{3}} + \mathbf{\frac{1}{4}}\ln(\mathbf{3})$$

23. Evaluate $\displaystyle\int_{\sqrt{2}}^2\frac{dx}{x^2\sqrt{x^2-1}}$**:**

From Problem 8 structure, antiderivative is $\frac{\sqrt{x^2-1}}{x}$.

$$\left[\frac{\sqrt{x^2-1}}{x}\right]_{\sqrt{2}}^2 = \frac{\sqrt{3}}{2}-\frac{1}{\sqrt{2}} = \mathbf{\frac{\sqrt{3}-\sqrt{2}}{2}}$$

24. Evaluate $\displaystyle\int_{\sqrt{2}}^2\frac{\sqrt{2x^2-4}}{x}\,dx$**:**

Factor out $\sqrt{2}$: $\sqrt{2}\int_{\sqrt{2}}^2\frac{\sqrt{x^2-2}}{x}\,dx$. Let $x=\sqrt{2}\sec\theta$. Antiderivative is $\sqrt{2x^2-4}-2\text{arcsec}\left(\frac{x}{\sqrt{2}}\right)$.

$$\left[\sqrt{2x^2-4}-2\text{arcsec}\left(\frac{x}{\sqrt{2}}\right)\right]_{\sqrt{2}}^2 = (\sqrt{4}-2\text{arcsec}(\sqrt{2}))-(0-0) = 2-2\left(\frac{\pi}{4}\right) = \mathbf{2}-\mathbf{\frac{\pi}{2}}$$

25. Evaluate $\displaystyle\int_1^3\frac{dx}{x^4\sqrt{x^2+3}}$**:**

Let $x=\sqrt{3}\tan\theta$. Limits: $x=1 \implies \theta=\pi/6$; $x=3 \implies \theta=\pi/3$.

$$\int_{\pi/6}^{\pi/3}\frac{\sqrt{3}\sec^2\theta\,d\theta}{9\tan^4\theta\cdot\sqrt{3}\sec\theta} = \frac{1}{9}\int_{\pi/6}^{\pi/3}\frac{\cos^3\theta}{\sin^4\theta}\,d\theta = \frac{1}{9}\int(1-\sin^2\theta)\sin^{-4}\theta\cos\theta\,d\theta$$

$$= \frac{1}{9}\left[-\frac{1}{3\sin^3\theta}+\frac{1}{\sin\theta}\right]_{\pi/6}^{\pi/3} = \mathbf{\frac{10\sqrt{3}}{243}}+\mathbf{\frac{2}{27}}$$

26. Evaluate $\displaystyle\int_0^3\frac{x^3}{(3+x^2)^{5/2}}\,dx$**:**

Let $u=3+x^2 \implies du=2x\,dx$. Limits: $x=0 \implies u=3$; $x=3 \implies u=12$.

$$\int_3^{12}\frac{u-3}{2u^{5/2}}\,du = \frac{1}{2}\int_3^{12}(u^{-3/2}-3u^{-5/2})\,du = \frac{1}{2}\left[-2u^{-1/2}+2u^{-3/2}\right]_3^{12}$$

$$= \left[\frac{1-u}{u^{3/2}}\right]_3^{12} = \frac{-11}{12\sqrt{12}}-\frac{-2}{3\sqrt{3}} = -\frac{11}{24\sqrt{3}}+\frac{16}{24\sqrt{3}} = \frac{5}{24\sqrt{3}} = \mathbf{\frac{5\sqrt{3}}{72}}$$

Part III: True-False & Concept Questions

27. False. For $\sqrt{x^2 - a^2}$, use $x = a\sec\theta$ (not $x = a\cos\theta$).

28. False. The restriction for $x = a\sin\theta$ is $-\pi/2 \leq \theta \leq \pi/2$.

29. True. For $\sqrt{a^2 - x^2}$, $x = a\sin\theta$ is the standard substitution.

30. False. The ellipse $x^2 + 16y^2 = 1 \implies \frac{x^2}{1^2} + \frac{y^2}{(1/4)^2} = 1$ has area $A = \pi ab = \pi(1)(1/4) = \pi/4 \neq 2\pi$.

31. $\displaystyle\int \frac{x}{x^2+9}\,dx$: Evaluating with $u = x^2+9$ yields $\frac{1}{2}\ln(x^2+9)+C$, which is faster than trig substitution.

32. $\displaystyle\int \frac{x^2}{x^2+4}\,dx$: Rewriting $\frac{x^2+4-4}{x^2+4} = 1 - \frac{4}{x^2+4}$ yields $x - 2\arctan(x/2) + C$ directly.